Halloween Night Hackathon

Five cursed problems. One endless night. Only ten survive till dawn.

NIGHT OF	DOORS	FINALE	EVALUATIONS
31 Oct 2026	8:00 PM	9:30 AM	3 rounds + final

01	🎃	**The Witch's Potion Pantry** Smart inventory & demand prediction	1st Year Only
02	👻	**The Midnight Phantom Hotline** Hinglish voice agent under 800 ms	AI + Systems
03	🧛	**The Possessed Butler** Prompt-injection-proof AI agent	AI Security
04	🧟	**The Midnight Crossroads** Indian traffic signal optimizer	CV + RL
05	💀	**The Coven of Five Hospitals** Federated learning across hospitals	Privacy + ML

Beware: at the stroke of midnight, every track will be cursed.

TRACK 01

The Witch's Potion Pantry

1st Year Only

AI + Full Stack + Web + Android

> *Madam Hexley runs the busiest potion shop in the village. Every Halloween she runs out of bat wings, mandrake root and ghost pepper at the worst moment, and angry customers turn into frogs. Build her a smart pantry.*

YOUR QUEST

- Track stock across her shelves: add, update and record what was sold.
- **Prophecy:** predict which ingredients will run out, and when.
- **Restock spell:** recommend how much of each ingredient to reorder.
- **Strange happenings:** flag unusual consumption, such as sudden spikes or missing stock.
- **Crystal ball:** charts of demand for each product over time.
- Build both a **web dashboard** and an **Android app** connected to the same backend.

SCORED ON

Prediction accuracy on hidden future sales data, anomalies caught, working web + Android app, and user experience.

MIDNIGHT CURSE

A full moon festival is announced for next week, and demand for 3 ingredients will triple. Your system must handle the sudden seasonal spike.

TRACK 02

The Midnight Phantom Hotline

AI + Systems

> *Ghosts keep calling the clinic and the haunted railway station, speaking a mix of Hindi and English, with chains rattling in the background and a signal weaker than a dying candle. Build an operator who never misses a call.*

YOUR QUEST

- A voice assistant that **books doctor appointments** or **answers railway queries** in code-mixed Hinglish.
- Handles **interruptions**, **background noise** and a **3G-level connection**.
- Replies within **800 ms end to end**, before the candle flickers.

SCORED ON

Task completion rate on hidden test calls, and latency.

WHY IT'S HARD

Streaming speech-to-text, turn-taking and the latency budget pull against each other.

MIDNIGHT CURSE

The latency limit drops to 600 ms, or callers start switching languages mid-sentence.



TRACK 03

The Possessed Butler

AI Security

> *Count Byteula's mansion is run by an AI butler who reads letters, browses the cursed web and sends messages for his master. Dark spirits have hidden curses in letters, web pages and scrolls, waiting to possess him.*

YOUR QUEST

- Build an LLM agent that **reads emails**, **browses web pages** and **takes actions** like sending mail or editing files.
- Protect it from possession: the organizers hide **prompt injections** in emails, web pages and PDFs that try to make it leak secrets or act against its master.

SCORED ON

Tasks completed minus successful possessions.

WHY IT'S HARD

There's no single known fix, so teams must design an architecture, not apply a patch.

MIDNIGHT CURSE

The Possession Round: teams write curses to attack each other's butlers, scoring points for both defence and attack.



TRACK 04

The Midnight Crossroads

Computer Vision + Reinforcement Learning

> *At the cursed crossroads, autos, bikes, buses, wandering cattle and the occasional zombie crawl with no lane discipline. Every red light lasts an eternity, and souls are stuck in traffic forever. Free them.*

YOUR QUEST

- **The All-Seeing Eye:** from CCTV-style video of mixed Indian traffic, estimate how many vehicles are queued at each approach.
- **The Signal Spell:** control the traffic lights in the SUMO simulator to cut average wait time.

SCORED ON

Reduction in average wait time on unseen junction scenarios.

WHY IT'S HARD

Real-world vision feeds a control loop, and test scenarios differ from training.

MIDNIGHT CURSE

A funeral procession (emergency-vehicle priority route) and a fog curse (low-visibility night footage) both hit the crossroads.

TRACK 05

The Coven of Five Hospitals

Privacy + Machine Learning

> *Five haunted hospitals each guard their patients' secrets. They must work together to diagnose a mysterious plague, but their oath forbids sharing records. Meanwhile the Soul Thief lurks, trying to discover whose records trained the model.*

YOUR QUEST

- Train a diagnostic model across **5 simulated hospitals** whose data is skewed differently.
- No raw data leaves any hospital (**federated learning**).
- Stay within a **differential-privacy budget**.
- Survive the Soul Thief's **membership-inference attacks**.

SCORED ON

Model accuracy, how much the attacks reveal, and communication cost.

MIDNIGHT CURSE

One hospital is corrupted and starts sending poisoned model updates. Detect it and stay accurate.

THE COVEN'S LAWS

Rules of the night

- All code must be written during the hackathon. Open-source libraries, pretrained models and public APIs are allowed; declare them in your README.
- Sample data is released at kickoff. Final scoring uses **hidden test sets** that no team sees in advance.
- The **Midnight Curse** is revealed at 12:00 AM and is part of Round 2 and Round 3 scoring.
- Teams must be present with a working build at every evaluation round. Missing a round scores zero for that round.
- **Code freeze at 6:45 AM.** Only the commit pushed before the freeze is evaluated.
- Attacking other teams' systems is allowed **only** in the official Track 3 Possession Round.
- The judges' decision is final.

THE NIGHT

Schedule of the damned

8:00 PM	🎃	**Opening ceremony** Track reveal, team check-in, costume parade
12:00 AM	🌑	**The Midnight Curse** Each track's twist is revealed
1:00 AM	👻	**Round 1 · The First Haunting** Architecture and baseline check
3:00 AM	🦇	**Witching-hour snack drop** Refuel. Dawn is still far away.
4:00 AM	🧛	**Round 2 · The Witching Hour Trial** Metrics on public test set, curse progress
6:45 AM	⚰️	**Code freeze** The graves are sealed
7:00 AM	💀	**Round 3 · The Graveyard Cut** Elimination round: hidden tests and live attacks
~8:30 AM	📜	**Top 10 announced** The survivors are named
9:30 AM	🔮	**Final · The Grand Séance** Top 10 teams present to the judges
~11:30 AM	👑	**Awards ceremony** The night ends

1:00 AM · ROUND 1

👻 The First Haunting

No elimination

Architecture and the basic flow working end to end.

4:00 AM · ROUND 2

🧛 The Witching Hour Trial

No elimination

Core metrics on the public test set and progress on the Midnight Curse. Live leaderboard published after.

7:00 AM · ROUND 3

💀 The Graveyard Cut

Elimination · Top 10 advance

Hidden test sets, live attacks (Tracks 3 & 5) and Midnight Curse handling.

9:30 AM · FINAL

🔮 The Grand Séance

Top 10 only

8 min presentation + live demo, then 4 min judge Q&A per team.

JUDGEMENT

How the living are judged

👻 Round 1 checklist (1:00 AM)

Track 1 · Pantry	Database and CRUD working; web and Android app connected to the backend.
Track 2 · Hotline	End-to-end voice pipeline answers at least one query.
Track 3 · Butler	Agent completes basic email, browsing and file tasks.
Track 4 · Crossroads	Vehicle detection running on the sample video.
Track 5 · Coven	Federated training loop running across all 5 hospitals.

SCORE WEIGHTS (CUMULATIVE)

15%	25%	60%
Round 1 · 1 AM	Round 2 · 4 AM	Round 3 · 7 AM

The **top 10 teams by cumulative score** after Round 3 advance to the Grand Séance. Final rankings are decided by the judges at the presentation round.

GLORY

Awards of the night

- 🧙 **Grand Sorcerer** — Overall winner
- 🌱 **Rising Witch** — Best first-year team (Track 1)
- 🛡️ **Exorcist** — Best defence in the Possession Round
- ⚡ **Speed Demon** — Lowest latency on the Phantom Hotline
- 💀 **Necromancer** — Best comeback after the Midnight Curse
- 🎭 **Best Costume** — Voted at the opening parade